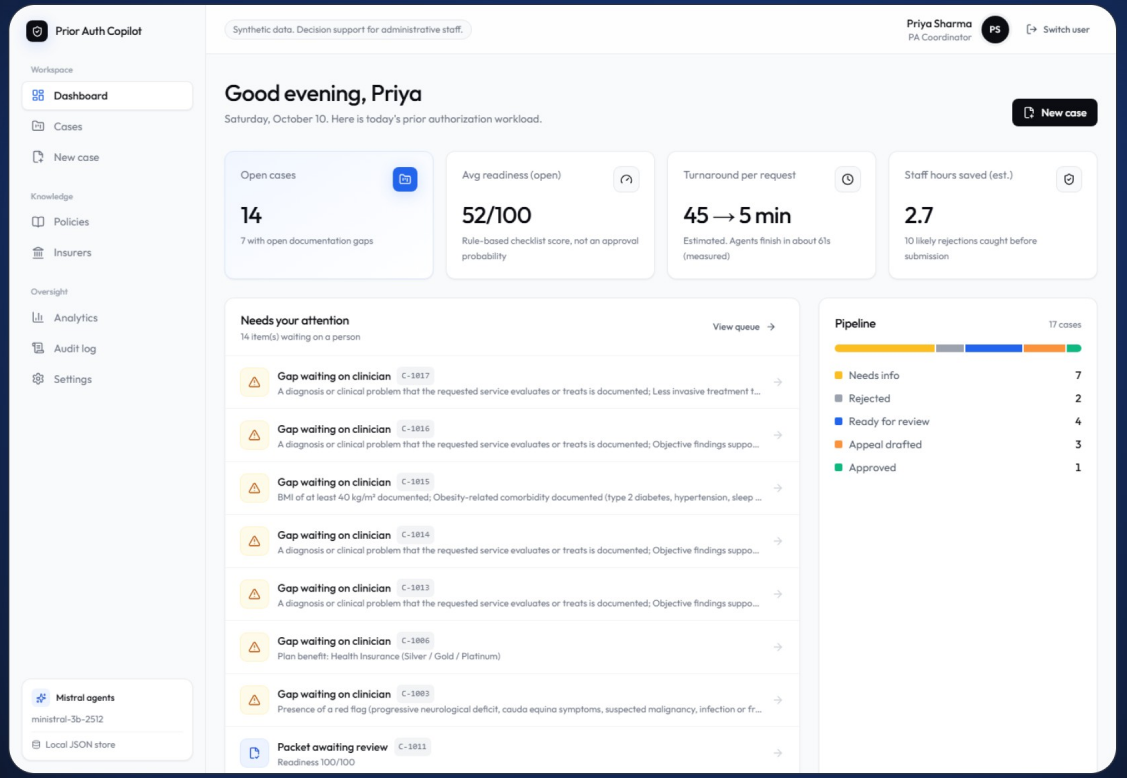
AICON'26 · BUILD WITH AI · HEALTH OPERATIONS

# Prior Auth Copilot

## Catch the denial before the insurer does.

AI agents that check every insurance pre-approval request against the insurer's own rules, with cited evidence, before it is submitted.

**Team Ctrl+Alt+Defeat**



THE PROBLEM

# Insurance pre-approval is slow, manual and easy to get wrong

1

### Read the chart

Notes, test results and history, by hand

2

### Find the insurer's rules

Different for every insurer and procedure

3

### Prove every rule

Match each requirement to evidence

4

### Write and submit

Letter of medical necessity and forms

⚠ One missing piece of proof means **rejection**, and the whole cycle starts again. Often the care was needed; the paperwork was incomplete.

THE REAL-WORLD PROBLEM

# A daily bottleneck for hospitals, here and abroad

### Hospital staff

An estimated 30 to 45 minutes of manual work per request

### Patients

Treatment waits while forms go back and forth

### Hospitals

Revenue lost to rejections that complete paperwork would prevent

**In Pakistan**, private and corporate health insurers also require pre-approval at panel hospitals for admissions and costly procedures, and the work is done by hand.

Time per request is an industry estimate, not our measurement.

DATA AND INPUT

# What the system works with

### Patient chart

Visit notes, therapy records, history and medications

### Reports and scans

Lab, ECG, echo, X-ray, angiography; PDF, photo or Word

### Insurer policies

72 policies, 9 insurers, incl. real Pakistani plan terms

### The order

Any procedure, from knee MRI to heart surgery, plus the insurer

All demo patients are synthetic. No real patient data is used.

WHERE AI IS USED

# A team of ten agents builds one packet

01 AI
### Report Reader
Reads PDFs, scans and photos

02 AI
### Intake
Pulls out facts, each with its quote

03 AI
### Policy Finder
Retrieves the insurer's own rules

04 AI
### Criteria Matcher
Met, missing or unclear, with proof

05 Code
### Gap-Closer
One clear question per gap

06 AI
### Letter Writer
Drafts the cited medical letter

07 AI
### Insurer Simulator
Argues like the insurer would

08 AI
### Verifier
Audits every citation

09 Person
### Human approval
Nothing is sent without it

10 AI
### Appeal Agent
Drafts the appeal if denied

Runs on Claude, Mistral, Groq or a local model. Works offline with a rule engine.

THE ROLE OF AI

# AI does the reading. Code keeps it honest.

## What the AI does

- Reads scanned reports and phone photos
- Extracts clinical facts with exact quotes
- Judges evidence against each insurer rule
- Writes the letter of medical necessity
- Argues as the insurer, then drafts appeals

## Guardrails we built

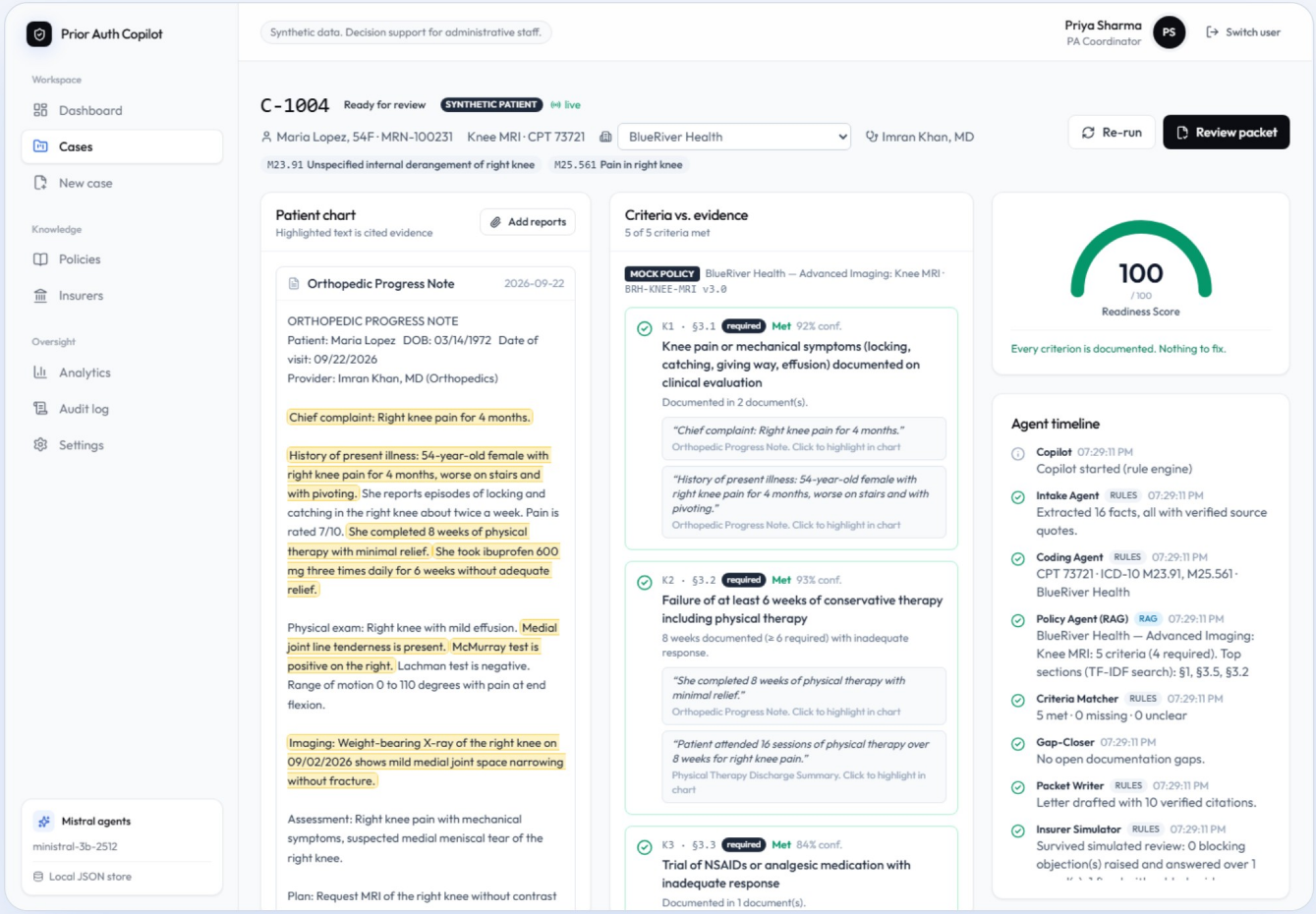
- Every claim must quote the chart word for word
- Non-medical or wrong-patient files are kept out
- If the AI disputes a match, a person decides
- Insurer objections must cite a real policy rule
- A person approves every submission; all logged

Decision support for administrative staff. It does not diagnose or order care.

WHAT THE USER GETS

# A ready, cited packet

- Checklist with each quote highlighted in the chart
- One clear question for each missing item
- Readiness score out of 100, with top fixes
- Cited letter of medical necessity, as PDF
- Simulated insurer review and audit report
- An appeal draft if the insurer still denies




Screenshot of the working app: case workspace

LIVE DEMO

# Same patient. Different insurer. Different answer.

1

New case: Hamza Qureshi, knee MRI, BlueRiver Health

2

Gap caught: 4 weeks of physio, the insurer needs 6. **Readiness 60**

3

Switch to Indus Health, which needs 4: every rule met. **Readiness 100**

4

Review the cited packet, approve, and see it in the audit log

No code changed between steps 2 and 3. Only the retrieved policy did.

Bonus: upload a scanned angiography report for a heart-surgery case

RESULTS, MEASURED

# Accurate, grounded and tested end to end

**30/30**
criteria verdicts correct

**100%**
of cited quotes found word for word in the chart

**5/5**
planted documentation gaps caught, 0 false alarms

**93**
automated end-to-end checks passing

**20** synthetic patients **19** procedures, plus any typed by name **9** insurers, **72** policies

Measured on 6 hand-labeled synthetic cases, with and without the Mistral reviewer. Not a clinical validation.

THE IMPACT

# How it improves the situation

| | Today | With Prior Auth Copilot |
|---|---|---|
| Missing proof | Found after the insurer rejects | Found before submission |
| Evidence | Searched for by hand | Every rule linked to an exact quote |
| New insurer | Learn its rules again | Policy retrieved; recheck in one click |
| Denials | Start the cycle over | Appeal drafted with counter-evidence |
| Trust | Hard to audit | Cited, verified, human-approved, logged |

Result: fewer preventable rejections, faster care, less staff time (time savings estimated until piloted).

# Every claim cited.
# Every decision human.

### Honest limits

- Synthetic patients and mock clinical policies
- Results measured on our own labeled cases
- Decision support only, not diagnosis

### Next steps

- Ingest real insurer policy documents
- Connect to hospital systems (HL7 FHIR)
- Pilot with a panel hospital in Pakistan

## Thank you

Team Ctrl+Alt+Defeat · Prior Auth Copilot